

AI POWERED REGULATORY COMPLIANCE CHECKER FOR CONTRACTS



Introduction

Organizations must constantly ensure their contracts align with ever-changing regulations.

The Challenge with Manual Compliance

Manual Tracking

Time-consuming, error-prone monitoring of regulatory updates

High Risk

Delays in detecting changes lead to legal penalties and non-compliance

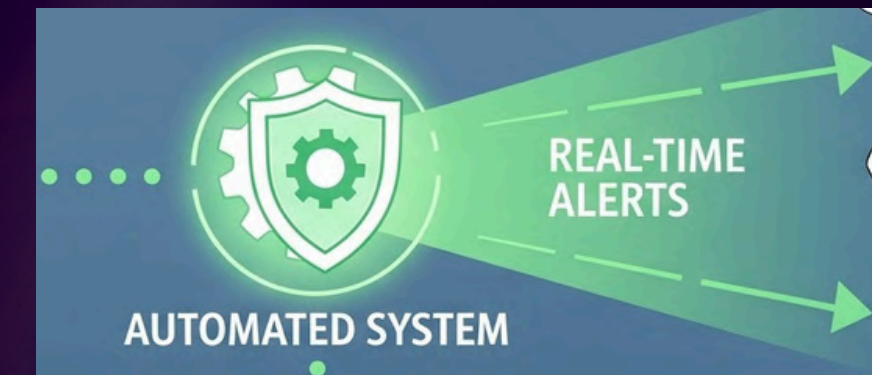
Resource Drain

Legal teams spend valuable time on manual compliance work



Solution

This project was created to build an automated system that addresses these challenges with proactive, real-time alerts.



Objective

The primary objective of this project is to create a robust system that:



Monitors Key Sources

Automatically scrapes and monitors official compliance-related websites and document templates for any changes or updates.



Provides Real-Time Notifications

Instantly notifies administrators and compliance teams about detected changes through integrated communication platforms.



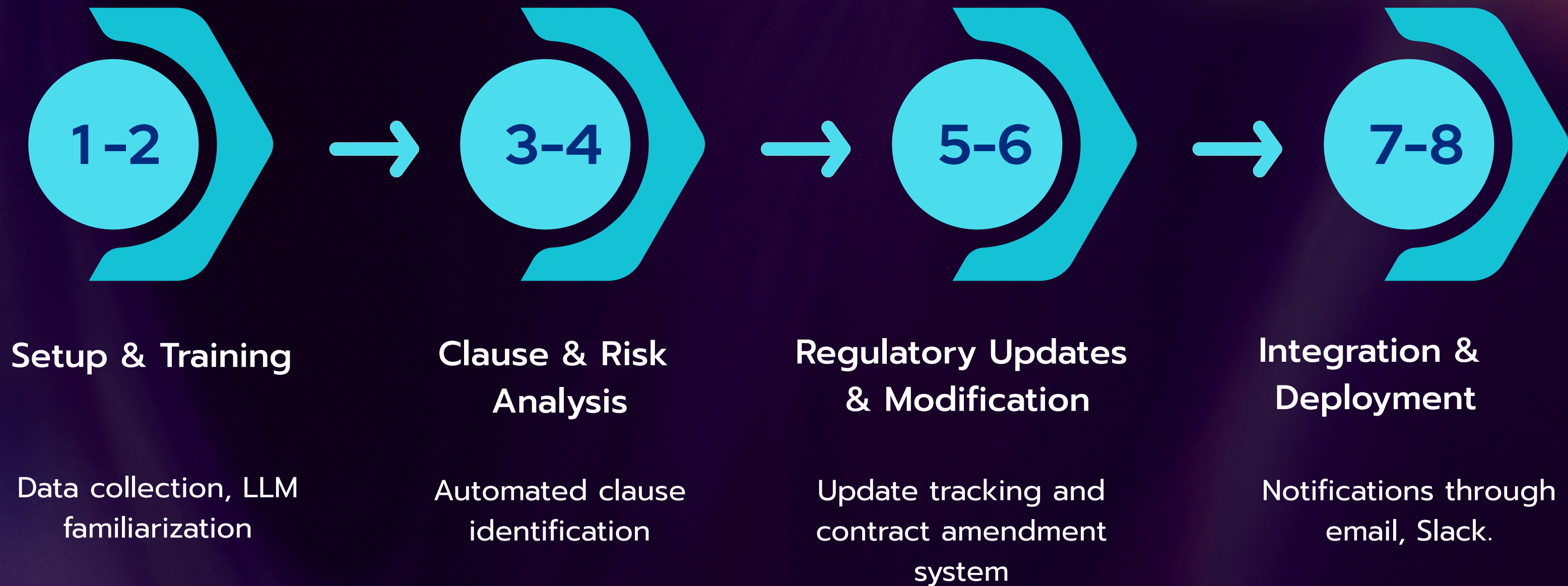
Supports Multiple Channels

Delivers alerts via:

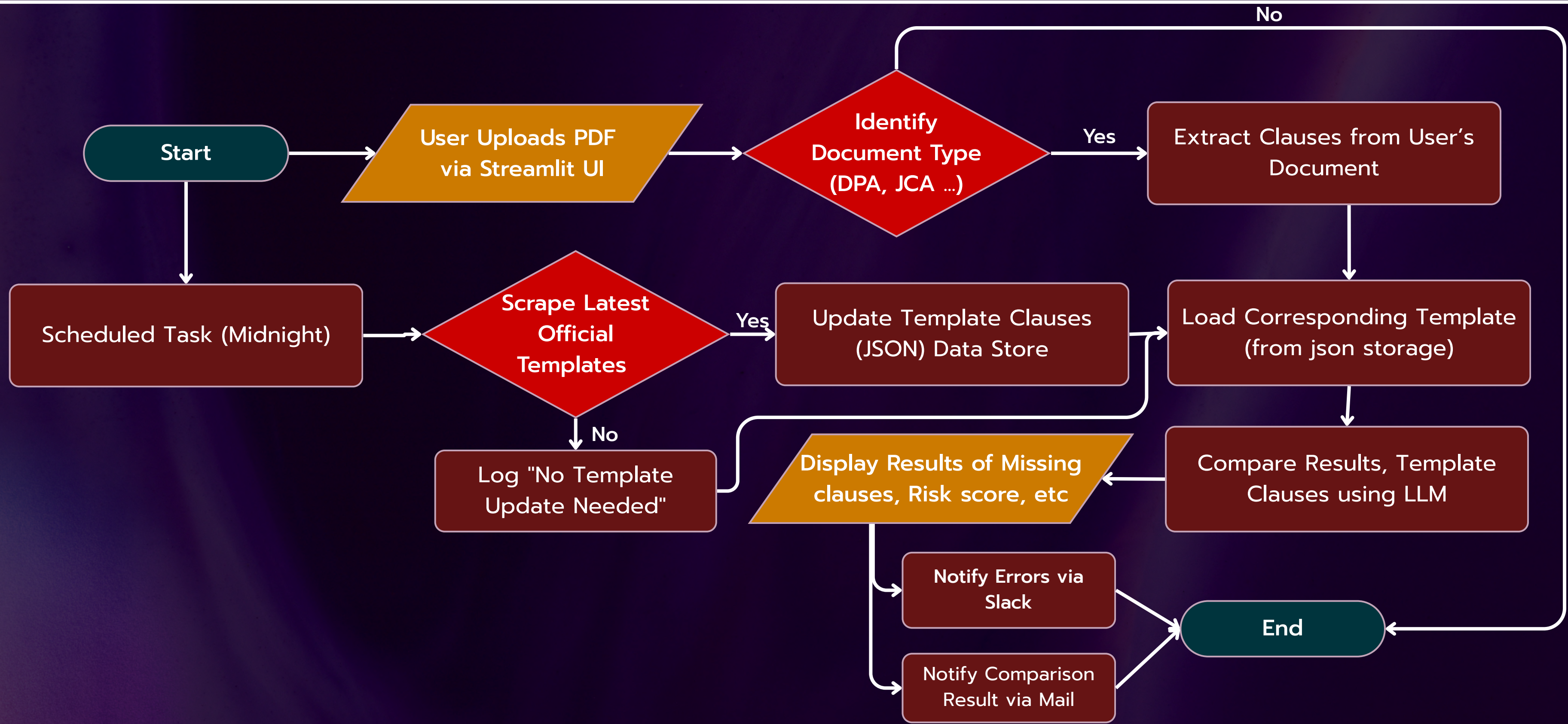
- Email (SMTP) for formal notifications.
- Slack for immediate, real-time team awareness.

Project Milestone

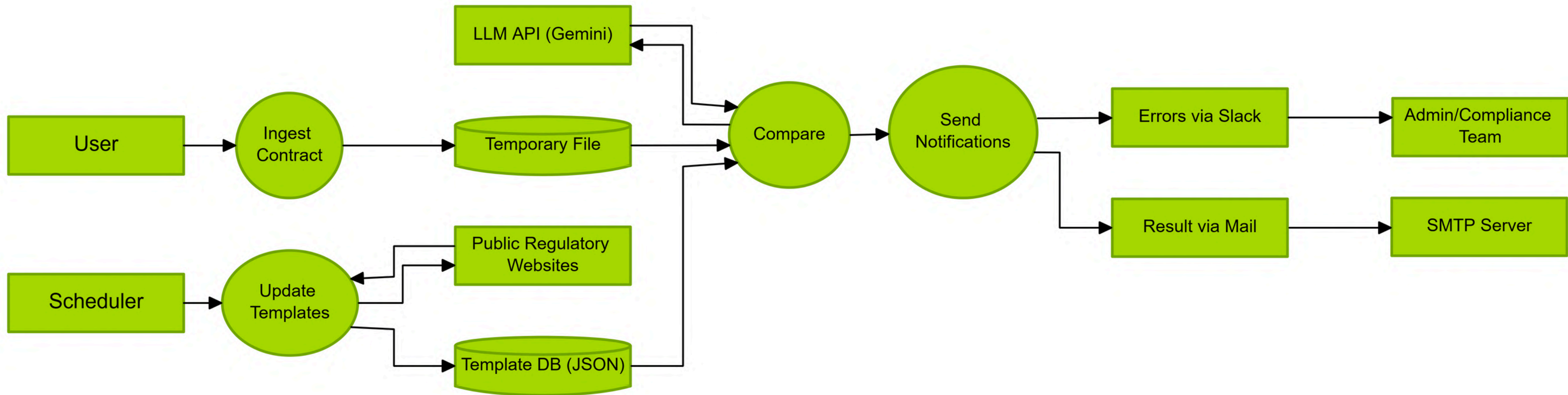
(Week wise)



FLOW CHART



Data Flow Diagram



System Architecture

Client Layer



User Interface (Frontend):
Streamlit-upload contracts



Notification Services: smtplib
for Email. slack_sdk for Slack
Webhooks.

Backend Layer



Scheduler: Triggers the
scraping process periodically



Backend (Core Logic): Python
Orchestrates the entire workflow



Scraping



Data Extraction



Comparison

Storage Layer



Users Uploaded PDFs



JSON Files (Template
Clauses)

External Layer



Gemini/Groq
APIs



Slack Webhook
API

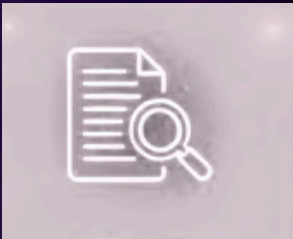


SMTP Server



Public
Websites

Modules Overview



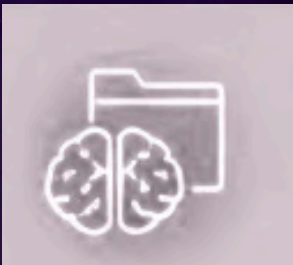
Clause Extraction
Module



Risk and
Compliance
Analysis



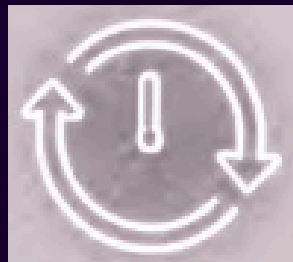
Real-time
Template
Scraper



Document Type
Detection
(AI-based)



Result Visualization
Module (Streamlit)



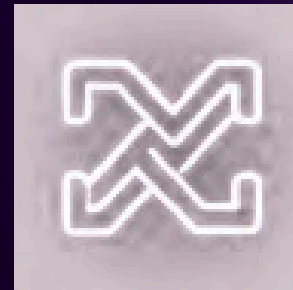
Automated
Scheduler



AI-based Clause
Extraction for
Unseen Files



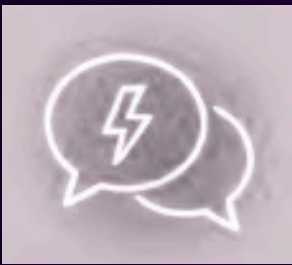
Notification System
(Email & Slack)



Threading & Parallel
Execution Module



Agreement
Comparison



Error Reporting
System (Slack Alerts)

Tech Stack

Programming Language

Python

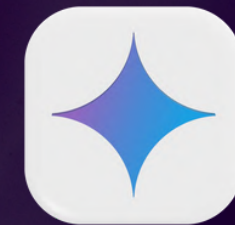


Core Libraries:

- **requests:** For making HTTP requests to download files.
- **schedule:** For automating the scraping jobs.
- **smtplib, email.mime:** For sending email notifications.
- **slack_sdk:** For sending Slack messages.
- **google-genai:** For interacting with Large Language Models.
- **PyPDF2:** For reading text from PDF files.
- **streamlit:** For building the interactive web UI.

APIs & Service

Gemini API, Groq API,
Smtplib API, Slack API



Development Tools

- **Visual Studio Code:** As the primary code editor.
- **GitHub:** For version control and code management.

Outcomes

Email Notification For Comparison Result

Dear [Recipient's Name],

Thank you for submitting the Joint Controller Agreement for review.

We've completed a detailed comparison against the GDPR-compliant template and identified several areas where the current contract deviates from best practices and regulatory expectations.

Comparison Result: - Missing Clauses: None. All clauses from the template are present in the new contract.

Potential Compliance Risks:

- Clause 1.1 (Definitions): Reference to "Article 27 of the GDPR" instead of "Article 26 of the GDPR" for the definition of "Agreement". Article 27 concerns representatives of controllers or processors not established in the Union, while Article 26 specifically addresses joint controllers. This is a factual error that could lead to misinterpretation of the agreement's legal basis.

- SCHEDULE 3: The table for "Data Processors" is missing the column heading "this Party" under "Data Processors engaged by". This is a minor formatting issue but could lead to ambiguity in identifying which party engaged which data processor.

- SCHEDULE 3: The table for "Transmission" is missing the column headings "this Party" under "Entity transmitting the Personal Data + country" and "this Party" under "Entity receiving the Personal Data + country". This is a minor formatting issue but could lead to ambiguity in identifying which party is transmitting or receiving data.

Risk Score (0-100): 30

Reasoning: The primary risk stems from the incorrect GDPR article reference in the "Definitions" clause (Clause 1.1).

Dismissing the foundational legal article for joint controllership (Article 27 instead of Article 26) is a significant error that could undermine the entire agreement's legal validity and interpretation regarding joint controller responsibilities. While the rest of the contract appears largely consistent, this fundamental error warrants a moderate risk score. The minor formatting issues in Schedule 3 are less critical but contribute to potential ambiguity.

Recommendations:

- Amend Clause 1.1 (Definitions) to correctly reference "Article 26 of the GDPR" instead of "Article 27 of the GDPR" for the definition of "Agreement".

User Interface



Contract Compliance Checker

Upload an agreement (PDF only)



Drag and drop file here

Limit 200MB per file • PDF

Browse files



(JCA) model-joint-controllership-agreement.pdf 0.8MB

.8MB



Processing your file...

Detected Document Type: Joint Controller Agreement

Clause Extraction Completed!!!



Comparison Result

- Missing Clauses:
 - Clause 1. heading is altered to "Clause 1. Definitions" in the new contract, but the clause_id is "3" instead of "Clause 1."
 - "LIST OF SCHEDULES" heading is present in the new contract, but the clause_id is missing.
 - "SCHEDULE 1:" heading is present in the new contract, but the clause_id is missing.
 - "SCHEDULE 2:" heading is present in the new contract, but the clause_id is missing.

- Missing or Incomplete Schedule Information: Although the headings for the schedules are present, the content within the schedules (e.g., specific details for application names, suppliers, responsibilities, data processing operations, security measures) is largely placeholder text. This means the contract, as provided, lacks the crucial operational details required for GDPR compliance in a joint controllership arrangement.

- Risk Score (0-100): 85

- Reasoning: The primary reason for the high-risk score is the incorrect GDPR article reference in Clause 1.1, which fundamentally misrepresents the legal basis for the joint controllership agreement. This error could lead to significant legal challenges regarding the validity and enforceability of the agreement's core purpose. Additionally, the lack of specific, filled-in details within the schedules means the contract is effectively a shell, failing to meet the transparency and accountability requirements of GDPR Article 26 regarding the respective responsibilities of joint controllers. The inconsistent clause numbering, while minor, points to a lack of rigorous review.

- **Recommendations:**

- **Correct GDPR Article Reference:** Immediately amend Clause 1.1 to refer to "Article 26 of the GDPR".
- **Complete All Schedules:** Ensure all schedules (1, 2, 3, and 4) are thoroughly completed with specific, detailed, and accurate information regarding the collaborative partnership, division of responsibilities, data processing operations, and technical/organizational security measures. This is critical for demonstrating GDPR compliance.
- **Standardize Clause Numbering:** Review and correct all clause IDs to align with the template's structure (e.g., "Clause 1.", "SCHEDULE 1:", etc.) for clarity and consistency.
- **Legal Review:** Conduct a comprehensive legal review of the entire document to ensure all provisions accurately reflect GDPR requirements for joint controllers and are legally sound.

Testing

Data Extraction Error

Contract Compliance Bot APP 7:46 PM
Test Slack Notification from Contract Compliance Bot

Error Occured in Template Data Extraction, Error is 404 NOT_FOUND. {'error': {'code': 404, 'message': 'models/gemini-1.5-flash is not found v1beta, or is not supported for generateContent. Call ListModels to see the list of available models and their support

Scraping Error

Error Occured in Scraping, Error is Invalid Elementary Object starting with b'\xab' @335865:

b'\x00\x00\x00\x00\x00\x00\x00\xd5\xe5\x8b\xbc\xfe\xaa\x0b\xd8\t\nzF+\xab+nCL\xa4\x9fQ\xefO\xea\x92\n\x0e]k\x1d|\x0f
o<\xfa\xb3\x82Z\x02O\xcftl\x9d:P#oG\xa0\xd6;\xf2n\xceV\x1d\n{\xeb\xc4\x9e*\xc4N\x02\x0e'

Downloading Error

Failed to Download File, Response status is 403 and Download Link is <https://www.thomsonreuters.com/content/dam/ewp-m/documents/thomsonreuters/en/pdf/global-sourcing-procurement/eu-eea-standard-contractual-clauses-v09-2021.pdf>

Future Enhancements

Interactive Compliance Dashboard:

A web-based dashboard to visualize compliance track changes, and manage historical versions of documents.

Multi-User Notification System:

Allow different users or teams to subscribe to alerts for specific types of documents

Integration with More Platforms:

Add support for sending notifications to other popular platforms like Microsoft Teams and Telegram.



Conclusion

This AI-Powered Regulatory Compliance Checker successfully automates a critical and time-consuming business process.



The project demonstrates the power of combining web scraping, AI, and automation to solve real-world business challenges effectively.

